

Identification of a Candidate Disease-Causing Variant in a Consanguineous Family with Intellectual Disability, Autism Spectrum Disorder, and Microcephaly Using Whole Exome Sequencing

A Whole Exome Sequencing (WES) Case Analysis Report

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Abstract

This report was prepared as part of an International Genomics Fellowship (July 1 – September 17, 2026), a collaboration between the Genomics Research Laboratory (Iran) and the Helix Institute of Biological and Research Sciences (USA), under the supervision of Dr. Sara Cheraghi.

This report presents the molecular genetic investigation of a consanguineous family (double first cousins) with two affected individuals (V:1 and V:4) presenting with developmental/intellectual delay, autism spectrum disorder, speech impairment, and progressive microcephaly (occipitofrontal circumference -2.77 SD at age 32 years in V:1, and -4.07 SD at age 30 years in V:4). Whole exome sequencing (WES) was performed and, following variant annotation and filtering, 55 candidate exonic variants across 55 genes remained. Given the consanguinity and the clinical presentation, an autosomal recessive inheritance pattern was assumed, and homozygous/hemizygous variants were prioritized. A rare, homozygous missense variant in *ALKBH8* (c.1966G>T; p.D656Y), predicted damaging by multiple in silico tools (CADD = 35), emerged as the leading candidate, consistent with the previously reported association between biallelic *ALKBH8* variants and autosomal recessive intellectual disability. Segregation analysis and Sanger confirmation are recommended as next steps.

1. Introduction

Intellectual disability (ID) combined with autism spectrum disorder (ASD), speech delay, and microcephaly is a clinically and genetically heterogeneous presentation. In consanguineous families, autosomal recessive (AR) inheritance is the most likely underlying mechanism, and homozygosity for rare, deleterious variants is the expected molecular signature. Whole exome sequencing (WES) has become the diagnostic method of choice for such presentations, allowing simultaneous interrogation of the coding regions of essentially all known genes.

The purpose of this report is to document the clinical and molecular work-up of a family with two affected individuals, describe the variant filtering strategy applied to WES data, and propose the most plausible causative variant based on population frequency, in silico pathogenicity predictions, ClinVar annotations, and consistency with the reported gene-disease relationship.

2. Case Presentation

The family originates from a consanguineous union (the parents are double first cousins — children of two sisters). The pedigree spans five generations (I–V) and includes two affected individuals in the youngest generation: V:1 (affected male) and V:4 (affected female). All other family members are reportedly unaffected.

2.1 Clinical Phenotype

- Developmental/intellectual delay in both affected individuals
- Autism spectrum disorder (ASD)
- Speech impairment / language delay

- Progressive microcephaly:
 - V:1 — head circumference at 32 years: -2.77 SD
 - V:4 — head circumference at 30 years: -4.07 SD

Both affected individuals are the offspring of a first-cousin marriage, strongly supporting an autosomal recessive mode of inheritance.

3. Materials and Methods

3.1 Sequencing

Genomic DNA was subjected to whole exome sequencing (WES). Sequencing reads were aligned to the human reference genome and variants were called following standard best-practice pipelines (alignment, duplicate marking, base quality score recalibration, and joint genotyping).

3.2 Annotation and Filtering

Called variants were annotated using ANNOVAR against RefGene, dbSNP, 1000 Genomes, ExAC, gnomAD-derived allele frequencies, ClinVar, and a panel of in silico pathogenicity predictors (SIFT, PolyPhen-2, MutationTaster, MutationAssessor, FATHMM, PROVEAN, CADD, and others). Variants were filtered to retain rare (population allele frequency < 1%), exonic, nonsynonymous variants. Given the consanguinity and clinical presentation, particular attention was given to variants present in the homozygous (or hemizygous, for X-linked loci) state, consistent with an autosomal recessive or X-linked recessive model.

4. Results

After filtering, 55 genes harboring rare, nonsynonymous exonic variants remained. Table 1 lists the three main candidate genes selected for detailed review, based on zygosity, in silico pathogenicity predictions, and known gene-disease associations.

Table 1. Main candidate genes selected for detailed review.

Gene	Chr	Protein Change	CADD	ClinVar	Zygosity
ALKBH8	Chr 11	p.D656Y	35	Not reported	Homozygous
WDR13	Chr X	p.G206W	32	Not reported	Homozygous
CHD8	Chr 14	p.R1895C	32	Not reported	Heterozygous

4.1 Prioritization Strategy

Of the 55 candidate genes, only four carried variants in the homozygous or hemizygous state: ALKBH8, WDR13 (X-linked), DMD (X-linked), and SYNE1. Among these, the DMD and SYNE1 variants are classified as Benign/Likely benign in ClinVar and were excluded. The WDR13 variant, while rare and predicted damaging, is hemizygous on chromosome X and lacks an established gene-disease relationship with intellectual disability or microcephaly; it remains a lower-priority candidate, relevant primarily if the sequenced individual is male.

4.2 Leading Candidate: ALKBH8

- Variant: chr11, c.1966G>T (NM_138775), p.Asp656Tyr (D656Y) — homozygous
- CADD Phred score: 35 (highly deleterious)
- In silico predictions: SIFT, PolyPhen-2, MutationTaster, FATHMM, and PROVEAN all classify this substitution as damaging/deleterious
- Population frequency: rare in gnomAD/ExAC ($\approx 0.02\%$ overall)
- ClinVar: currently uncertain significance / not previously classified
- Gene function: ALKBH8 encodes a tRNA-modifying enzyme (wobble uridine hydroxylation) important for translational fidelity in neurons
- Gene-disease association: biallelic loss-of-function or damaging missense variants in ALKBH8 have been reported in autosomal recessive intellectual disability (MRT71), often accompanied by speech delay and, in a subset of patients, microcephaly

5. Discussion

The homozygous ALKBH8 variant identified in this family is the strongest candidate to explain the observed phenotype for several converging reasons: (1) it is compatible with the autosomal recessive inheritance pattern expected in a consanguineous pedigree; (2) it is rare in population databases; (3) it is predicted deleterious by essentially all computational pathogenicity tools; and (4) ALKBH8 has a previously reported, phenotypically consistent association with intellectual disability, speech delay, and, in some patients, microcephaly.

Other heterozygous variants identified in the filtered list — including well-known neurodevelopmental genes such as CHD8 — do not fit the recessive inheritance model implied by the consanguinity and by the fact that two siblings are affected while parents are presumably unaffected carriers. CHD8-related disorders, in particular, are typically caused by de novo dominant loss-of-function variants and are often associated with macrocephaly rather than microcephaly, which further argues against CHD8 as the causal gene in this family.

5.1 Limitations

- The candidate variant has not yet been confirmed by Sanger sequencing
- Segregation analysis in additional family members (parents as obligate carriers, unaffected siblings) has not been performed

- WES data from the second affected individual (V:4) was not available for direct comparison at the time of this analysis
- Functional/biochemical validation of the ALKBH8 p.D656Y variant has not been performed

6. Conclusion and Recommendations

A rare, homozygous, predicted-damaging missense variant in ALKBH8 (p.D656Y) is proposed as the leading candidate for the phenotype observed in this consanguineous family. The following steps are recommended to confirm this finding:

- Sanger sequencing confirmation of the ALKBH8 variant in the proband
- Segregation analysis across the family (parents, unaffected siblings, and the second affected individual V:4)
- Re-sequencing or targeted analysis of V:4 to confirm the presence of the same homozygous variant
- Literature review and, if feasible, functional studies to further support pathogenicity
- Genetic counseling for the family regarding autosomal recessive inheritance and recurrence risk

7. References

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